

Technical Supplement B

Supernova Compression, Residual Localization, and Public-Lineage Limits in the HODN Network

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Scope: synthesis of four frozen audit packages. No new H_0 estimate, causal explanation, DOI, repository record, or endorsement is asserted.

B1. Purpose and scope

This supplement documents the frozen 277-row Type Ia supernova block audited in the HODN network, the exact fixed-covariance, one-intercept linear Gaussian compression identity, the residual information lost by that compression, the localization of an unexpectedly low same-name contrast statistic, and the boundary of public provenance work. It distinguishes three questions:

1. Is the scalar compression sufficient for the declared network target?
2. Does the compression retain enough information to assess residual adequacy and pattern?
3. Can public products connect candidate photometric inputs and configurations to the exact executed path that generated each final corrected row?

The answers in the frozen record are, respectively: yes for the fixed-covariance, one-intercept linear Gaussian target; no for residual diagnostics; and partial but not complete for executed lineage. No claim is made for additional slopes, nuisance parameters, covariance estimation, alternative likelihoods, or model selection. No row-deletion rule, covariance correction, survey cause, BBC overcorrection, or replacement H_0 is proposed.

B2. Frozen data block and model

Let $d \in \mathbb{R}^{(n)}$ denote the Hubble-flow supernova vector with $n=277$, and let $C \in \mathbb{R}^{(277 \times 277)}$ be its frozen positive-definite covariance within this branch. The covariance is treated as fixed; no covariance hyperparameters, nuisance slopes, or additional mean terms are fitted. The declared linear Gaussian model contains one intercept a :

$$d = a\mathbf{1} + \varepsilon, \varepsilon \sim N(0, C).$$

Define

$$s = \mathbf{1}^T C^{-1} \mathbf{1}, t = \mathbf{1}^T C^{-1} d.$$

The generalized least-squares estimate and its variance are

$$\hat{a} = (t)/(s), V = (1)/(s).$$

The frozen data give

$$\hat{a} = 0.7163834210954622, \sqrt{V} = 0.0018926416391806472.$$

Embedding the full 277-row block or the scalar ($\hat{a}, V$) into the frozen network produced the same target estimate and covariance at the registered numerical precision. The network baseline remained

$$H_0 = 73.49875364360662 \pm 0.8088000253378453 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

B3. Complete-square identity

The full quadratic form is

$$Q(a) = (d - a1)^T C^{-1} (d - a1).$$

Expanding,

$$Q(a) = d^T C^{-1} d - 2at + a^2 s.$$

Since $\hat{a} = t/s$, completing the square gives

$$Q(a) = (d^T C^{-1} d - (t^2)/(s)) + s(a - \hat{a})^2.$$

Define

$$\chi^2_{\text{(min)}} = d^T C^{-1} d - (t^2)/(s).$$

Then

$$Q(a) = \chi^2_{\text{(min)}} + ((a - \hat{a})^2)/(V).$$

The Gaussian likelihood factorizes as

$$L(a | d, C) \propto \exp(-(1)/(2)\chi^2_{\text{(min)}}) \exp[-(1)/(2)((a - \hat{a})^2)/(V)].$$

The first factor does not depend on a ; the second contains all likelihood dependence on the one target. Therefore ($\hat{a}, V$) is sufficient for inference on a under this fixed-covariance, one-intercept linear Gaussian model. It is not asserted to be sufficient after adding nuisance parameters, estimating the covariance, or changing the likelihood. This is a target-specific sufficiency statement in the sense of Fisher and later measure-theoretic formulations [B1, B2].

B4. What the scalar omits

For the frozen 277-row block,

$$\chi^2_{\text{(min)}} = 206.7606364373241$$

for 276 residual degrees of freedom. The scalar pair ($\hat{a}, V$) does not contain this constant. It also does not contain the direction of the whitened residual vector and therefore cannot support subspace, pattern, or influence diagnostics.

The final internal validation recomputed the identity using a separate parser and Cholesky and eigendecomposition routes. Across seven profile values of a , the maximum discrepancy between the expanded quadratic form and the complete-square expression was

$$3.410605131648481 \times 10^{-13}.$$

This confirms that the omitted term is not a numerical approximation introduced by the scalarization; it is the exact parameter-independent remainder of the frozen model.

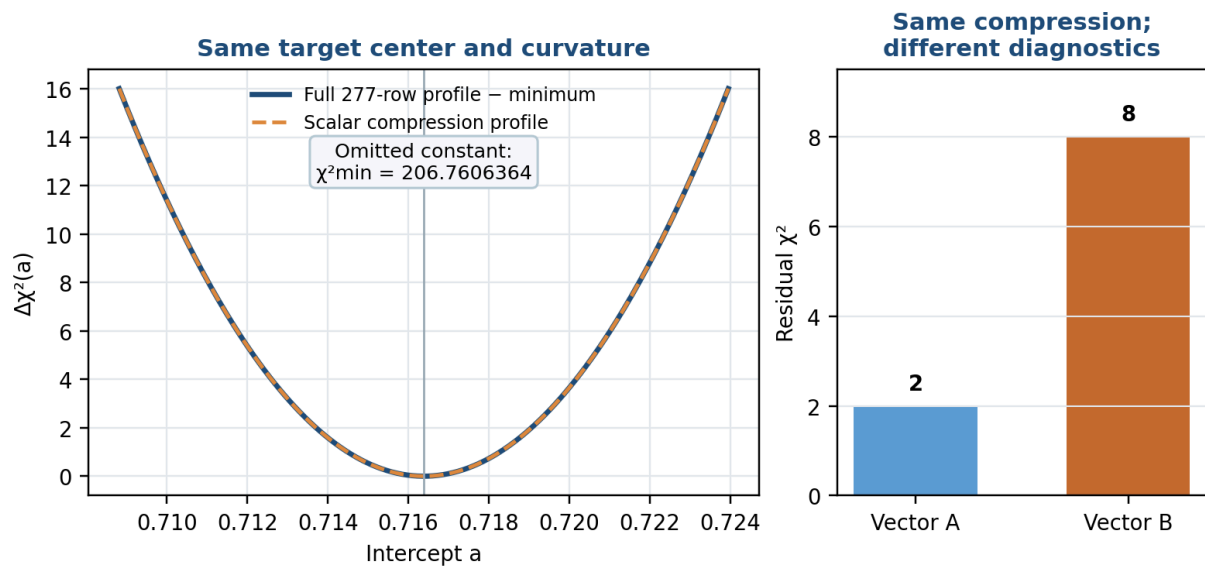


Figure B1. Parameter and diagnostic sufficiency are different. Left: the full and compressed one-intercept likelihoods have the same center and curvature; their negative log-likelihoods differ only by the omitted residual constant. Right: exact synthetic vectors can share the same compressed mean and variance while having residual chi-squares 2 and 8.

B4.1 Exact equal-compression, unequal-residual fixture

The final package included exact synthetic vectors with the same sample mean and the same declared compression variance but different residual patterns. Their residual chi-squares were exactly 2 and 8. Both generate the same target-dependent quadratic term, so any inference based only on the compressed scalar is identical. Yet their fit to the one-intercept pattern differs by a factor of four in residual energy.

The fixture proves a logical point independent of the HODN numbers: parameter sufficiency does not entail diagnostic sufficiency. Residual analysis has long been treated as a separate component of model assessment [B3,B4], and predictive discrepancies likewise depend on features that a target-only compression may discard [B5].

B5. Orthogonal residual decomposition

The audit next asked whether the residual energy was distributed uniformly across scientifically meaningful subspaces. Thirty supernova names appeared in multiple rows. These groups contained 69 rows. For group g with $m(g)$ rows, the within-name contrast subspace has dimension $m(g)-1$. Summing over the 30 groups gives

$$\sum_g (m(g)-1) = 69 - 30 = 39$$

within-name contrast degrees of freedom. The remaining between-name residual subspace has dimension

$$276 - 39 = 237.$$

Using the covariance-aware orthogonal decomposition registered in Phase 1A, the total residual split as follows.

Table B1. Residual localization in the frozen one-intercept model.

Component	χ^2	Degrees of freedom	Share of total residual
Same-name/multirow contrasts	11.209315063602752	39	5.42%
Between-name modes	195.55132137369947	237	94.58%
Total	206.760636437324	276	100.00%

The within-name statistic is unusually low under the registered chi-square-family calibration. The branch used a beta-ratio tail calculation for the partition and obtained lower-tail probability

$$9.368362232281232\times10^{-5}$$

and two-sided probability

$$1.8736724464562464\times10^{-4}.$$

Here, the beta-ratio probability is conditional on the registered residual partition. It is distinct from the Phase 1C marginal lower-tail probability of $3.6795245876638087\times10^{-6}$ under a χ^2 distribution with 39 degrees of freedom; the two quantities answer different reference questions and are not numerically inconsistent.

The statistic is a residual localization result. It does not state that the rows are duplicates, that their errors are dependent, or that their covariance is overestimated.

B6. Covariance robustness within the frozen subspace

Phase 1C applied the same 39-dimensional contrast question to registered covariance variants. The principal results were:

Table B2. Same-name contrast statistic under frozen covariance variants.

Variant	χ^2	df	Registered lower-tail or two-sided interpretation
Full frozen covariance	11.209315063602716	39	Low; canonical result
No rowwise velocity term	14.734236	39	Low flag persists
STATONLY covariance	16.233447508593247	39	$p=4.856832550848106\times10^{-4}$ under registered tail rule

The canonical package also checked an alternate contrast/null basis, 32 orthogonal rotations across five baselines, and a separate project-internal implementation using an independently written parser and eigendecomposition. The final closure audit reran the canonical Phase 1C verifier: 24/24 gates passed, and the separate verifier passed.

These checks make a basis-selection or single-code-path explanation unlikely within the frozen statistical question. They do not test a new covariance model or provide an external replication.

B7. Identity and covariance mapping

Before causal interpretation, the audit verified that the statistical objects were correctly aligned:

- 277 of 277 HODN supernova rows mapped to official Hubble-flow entries under the registered identifiers;
- all 76,729 covariance elements, 277^2 , were recovered in the expected order;
- 30 multirow groups comprising 69 rows were identified; and
- the same fixed one-intercept/covariance contract was used across compression and residual analyses.

These checks close artifact identity for the audited statistical block. They do not establish the production history of each corrected magnitude.

B8. Public candidate-input provenance

The public-lineage analysis was conducted in stages to avoid treating candidate filenames as executed provenance.

B8.1 Phase 1D: direct candidate search

Seven frozen public directories containing 847 eligible files were searched using the fixed CID and survey crosswalk. Unique compatible public photometric input candidates were found for 38 of the 69 multirow rows. Thirty-one rows were unresolved under that initial crosswalk. Among the 48 registered cross-series row pairs, ten had both sides resolved at this stage, and none had byte-identical observation lines.

B8.2 Phase 1E: target-excluded survey bridge

The unresolved 31 rows belonged to survey codes 51, 57, and 65. A public-internal crosswalk was constructed from eligible rows while excluding the target CID from the inference set. Sixty-two anchors from 74 eligible rows supported all three survey-code mappings. Under that non-circular rule, all 31 unresolved rows connected to unique compatible candidates within the frozen seven-directory universe.

Combining Phases 1D and 1E gave candidate coverage for 69/69 rows. “Candidate coverage” means that a unique compatible public input was identified under the registered rules; it does not mean that an executed production manifest names that file.

B8.3 Phase 1F: observation, filter, and configuration layers

The 48 registered cross-series pairs were compared at the observation-payload layer:

- 0/48 pairs showed byte-exact reuse of observation lines;
- 4/48 met the registered single-rounding numeric-compatibility rule;
- 434/434 row-filter records mapped to public KCOR text or transmission assets;
- 6,744/6,744 observations were covered by those mappings; and
- 7/7 data series had configuration-level pipeline anchors.

The negative byte-exact result rules out one narrow form of public file reuse under the frozen comparison. It does not prove statistical independence, because separate files can derive from shared exposures, calibrations, templates, host-subtraction products, or processing stages. The four numeric-compatible pairs are likewise not proof of shared raw observations; they only satisfy the preregistered payload rule.

B9. The executed-lineage frontier

The public record did not close the following chain for each final row:

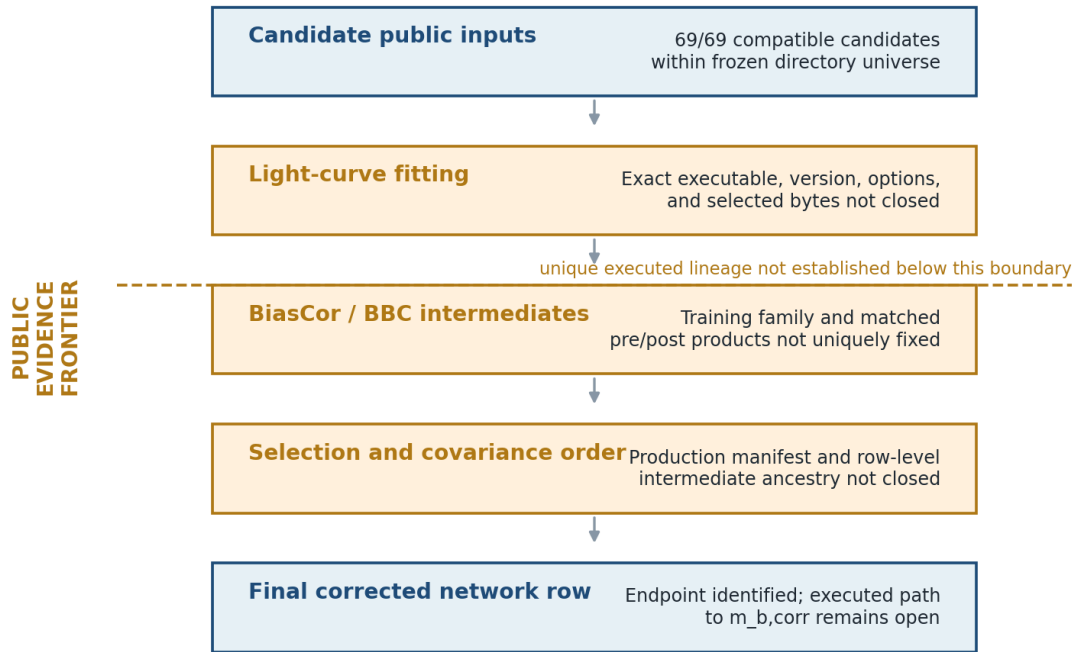


Figure B2. Executed-lineage frontier. Public products support candidate inputs, filters, and configuration anchors, but the audit did not recover a unique manifest connecting exact input bytes and executable options through BBC intermediates to each final corrected network row.

Repository templates and path families show how a pipeline may be configured, but unresolved environment variables and multiple candidate families prevent a unique reconstruction of the actual run. Final-vector fields demonstrate the endpoint but not the exact executed ancestry. For this reason, the branch state is a public-evidence frontier rather than a causal closure.

B10. Relation to the Pantheon+ BBC source-readiness audit

The separate BBC source-readiness branch fixed the Pantheon+ DataRelease repository at commit `c447f0fea703fcd0fff57de5000947b5ca81286b`. It confirmed public final vectors, corrected-magnitude fields, covariance products, ten data-sized simulations, and multiple PIPPIN BiasCor families. It did not find a uniquely versioned public bundle linking BiasCor truth, matched fitted outputs, BBC pre/post vectors, the corresponding covariance, and a fully resolved run configuration.

This upstream HOLD is consistent with the same-CID frontier. The residual statistic can be computed and localized from final statistical products, while the executed generative path required to explain it remains open. Neither result refutes the BBC method as described by Kessler and Scolnic [B6] or the published Pantheon+ analyses [B7–B9].

B11. What evidence would justify re-entry

The highest-value re-entry object is not another contrast basis applied to the same 277 rows. The canonical package already includes alternate-basis and orthogonal-invariance checks. Re-entry would be justified by one or more of the following:

- a versioned production manifest naming the exact input files, checksums, executables, commits, configuration options, and environment;
- matched light-curve-fit outputs and BiasCor/BBC intermediate products for the final rows;
- a row-level ancestry table from the final corrected-magnitude field to its executed inputs;
- an official truth→fit→pre/post→covariance closure bundle;
- a documented covariance-generating model for same-object or shared-calibration dependencies; or
- an external replication or concrete counterexample that changes the frozen residual claim.

Absent such evidence, further internal variation of the same statistic has low expected information gain.

B12. Interpretation and nonclaims

Under the frozen fixed-covariance, one-intercept linear Gaussian model, the compression preserves the parameter-dependent likelihood and network target but discards the minimized residual and its directional structure. The same-name contrast subspace contains an unexpectedly low fraction of residual energy, and that localization is internally robust to the registered basis and covariance checks. Public files support extensive candidate, filter, and configuration mapping, but not exact executed final-row ancestry.

These results do not establish:

- duplicate observations or byte reuse beyond the tested files;
- statistical independence or dependence of any named pair;
- an overestimated or incorrect covariance;
- a survey-specific, calibration, photometry, BBC, or pipeline cause;
- which row, if any, should be removed or changed;
- that Pantheon+ BBC corrections are excessive or invalid;
- a corrected supernova intercept or Hubble constant; or
- external replication of the project's internal checks.

B13. Bounded conclusion

For the fixed-covariance, one-intercept linear Gaussian HODN supernova model, scalar compression is exact for the declared target but insufficient for model diagnostics. The omitted residual term is 206.7606364373241 for 276 degrees of freedom, and an unusually low component is localized to 39 same-name contrast degrees of freedom. The localization survives the registered internal robustness checks. Public provenance reaches candidate inputs, filters, and configuration anchors but stops before executed run-to-final-row lineage. The scientific result is therefore a robust diagnostic location plus an explicit provenance HOLD, not a causal explanation or correction.

References for Supplement B

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